



Universitat
Pompeu Fabra
Barcelona

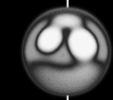
Wikidata Vandalism Detection with Graph-Linguistic Fusion

Authors:

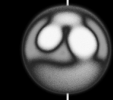
Mykola Trokhymovych

Diego Sáez Trumper

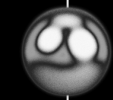
Introduction



Wikidata is the largest open-source, multilingual knowledge graph.



Wikidata content is frequently used to power other websites and products.



Wikidata has, on average, around **8 edits per second**.
Damaging edits should be identified and reverted.

Content representation challenge



- **Background:** The wikitext representation of a Wikidata item is a complex nested structure of dictionaries or lists.
- **Problem:** Content change parsing is complicated, as it can consist of structure change (e.g., single value to list) and value changes of different types (e.g., text in any language, numeric value, date, etc.).
- **Impact:** Previous iterations of models aiming to detect vandalism in Wikidata used very limited signals from content changes themselves.

```

{"type":"item","id":"Q105644691","labels":{"es":{"language":"es","value":"Eva Mora"},"nl":{"language":"nl","value":"Eva Mora"},"en":{"language":"en","value":"Eva Mora"},"ca":{"language":"ca","value":"Eva Mora"},"gl":{"language":"gl","value":"Eva Mora"}},
"descriptions":{"es":{"language":"es","value":"periodista y reportera espa\u00f1ola"},"en":{"language":"en","value":"Spanish journalist and reporter"},"ca":{"language":"ca","value":"periodista i reportera espanyola"},"nl":{"language":"nl","value":"Spaanse journalist en verslaggever"},"gl":{"language":"gl","value":"periodista e reporteira espa\u00f1ola"}},
"aliases":[],"claims":{"P31":[{"mainsnak":{"snaktype":"value","property":"P31","hash":"ad7d38a03cdd40cdc373de0dc4e7b7fcbccb31d9","datavalue":{"value":{"entity-type":"item","numeric-id":5,"id":"Q5"},"type":"wikibase-entityid"}},
"type":"statement","id":"Q105644691$c2de9533-4602-0ec1-2a43-87b3951f3c29","rank":"normal"}],
"P21":[{"mainsnak":{"snaktype":"value","property":"P21","hash":"5760796ff6ebc63aae12cdcbf509b07ebf0bd201","datavalue":{"value":{"entity-type":"item","numeric-id":6581072,"id":"Q6581072"},"type":"wikibase-entityid"}},
"type":"statement","id":"Q105644691$8558b1cc-4b29-6a19-5d75-8ec1eec2160b","rank":"normal"}],
"P106":[{"mainsnak":{"snaktype":"value","property":"P106","hash":"943348a1c561d670ec8f5fab1132465a7c779b55","datavalue":{"value":{"entity-type":"item","numeric-id":1930187,"id":"Q1930187"},"type":"wikibase-entityid"}},
"type":"statement","id":"Q105644691$72d4d34d-4249-00f6-ad4b-57d541537e0a","rank":"normal"},
{"mainsnak":{"snaktype":"value","property":"P106","hash":"54bde886150c5999f15b4c922af3ad481c1fb3f9","datavalue":{"value":{"entity-type":"item","numeric-id":42909,"id":"Q42909"},"type":"wikibase-entityid"}},
"type":"statement","id":"Q105644691$bd793745-4867-4e4d-559d-acd956b0925a","rank":"normal"}],
"P569":[{"mainsnak":{"snaktype":"value","property":"P569","hash":"a9ad62614bf5fe93c6fe7aa0b91d622711fb48ff","datavalue":{"value":{"time":"+1974-10-21T00:00:00Z","timezone":0,"before":0,"after":0,"precision":11,"calendarmodel":"http://www.wikidata.org/entity/Q1985727"},"type":"time"}},
"type":"statement","id":"Q105644691$da180ff2-403b-2016-fb21-1c72f75c3724","rank":"normal"}],
"P19":[{"mainsnak":{"snaktype":"value","property":"P19","hash":"47b32c1cefda5f02a42335eadda1c2326890a691","datavalue":{"value":{"entity-type":"item","numeric-id":2807,"id":"Q2807"},"type":"wikibase-entityid"}},
"type":"statement","qualifiers":{"P131":[{"snaktype":"value","property":"P131","hash":"936ab6c53e8d205edfe58fedaaaf3d89726273efa","datavalue":{"value":{"entity-type":"item","numeric-id":5756,"id":"Q5756"},"type":"wikibase-entityid"}]}],
"qualifiers-order":["P131"],"id":"Q105644691$03e679eb-409c-55d5-6668-9996fe8c5105","rank":"normal"}],
"P69":[{"mainsnak":{"snaktype":"value","property":"P69","hash":"42e162280ea49a1307fa954d567cd01e3369306e","datavalue":{"value":{"entity-type":"item","numeric-id":1247135,"id":"Q1247135"},"type":"wikibase-entityid"}},
"type":"statement","qualifiers":{"P512":[{"snaktype":"value","property":"P512","hash":"404b296142d007467ba330f0e333955b84ddb484","datavalue":{"value":{"entity-type":"item","numeric-id":43236300,"id":"Q43236300"},"type":"wikibase-entityid"}]}],
"P580":[{"snaktype":"value","property":"P580","hash":"f77e401c3294559f5f96c8fc18f16d1c6aad6e22","datavalue":{"value":{"time":"+1998-00-00T00:00:00Z","timezone":0,"before":0,"after":0,"precision":9,"calendarmodel":"http://www.wikidata.org/entity/Q1985727"},"type":"time"}]}],
"P582":[{"snaktype":"value","property":"P582","hash":"f84ab79307da73e0195993b8bfe549b6607b374f","datavalue":{"value":{"time":"+2002-00-00T00:00:00Z","timezone":0,"before":0,"after":0,"precision":9,"calendarmodel":"http://www.wikidata.org/entity/Q1985727"},"type":"time"}]}],
"qualifiers-order":["P512","P580","P582"],"id":"Q105644691$a344324c-4322-9409-07ba-e76b06735e1e","rank":"normal"},
{"mainsnak":{"snaktype":"value","property":"P69","hash":"5d67c53bf40bb20884648f285a1a4cea6cd00fe8","datavalue":{"value":{"entity-type":"item","numeric-id":5854722,"id":"Q5854722"},"type":"wikibase-entityid"}},
"type":"statement","qualifiers":{"P512":[{"snaktype":"value","property":"P512","hash":"7a8ce79aa6d1b7406d20d9af185477a31326031e","datavalue":{"value":{"entity-type":"item","numeric-id":45892697,"id":"Q45892697"},"type":"wikibase-entityid"}]}],
"P580":[{"snaktype":"value","property":"P580","hash":"7dc2821fe3f7d0644f5ada26

```

Fairness challenge

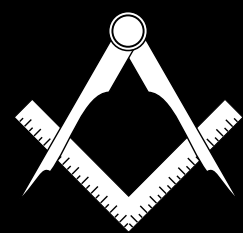


- **Background:** Anonymous editors usually have a higher revert rate
- **Problem:** Models overfit, causing anonymous editors discrimination
- **Impact:** New/anonymous editors are not converting to active editors, so their number decreases in the long term perspective.

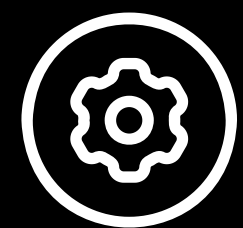
Goal & contribution

- **Goal:** Create a model to help Wikidata editors to identify edits that require patrolling.

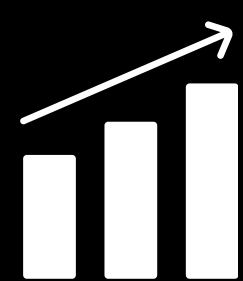
- **Contributions:**



- Defining the architecture of a new open-source, multilingual system for vandalism detection in Wikidata



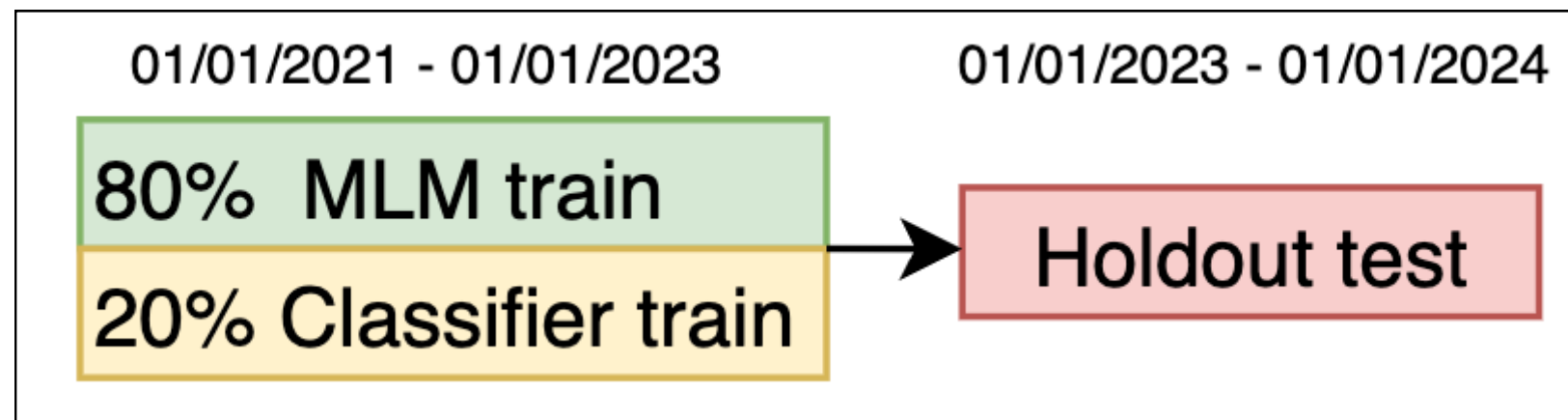
- Presenting a methodology for mining fine-grained content change features.



- Demonstrating preliminary results of significant model improvement compared to the current production model in terms of accuracy and fairness.

Data preparation & processing

Train-test split logic

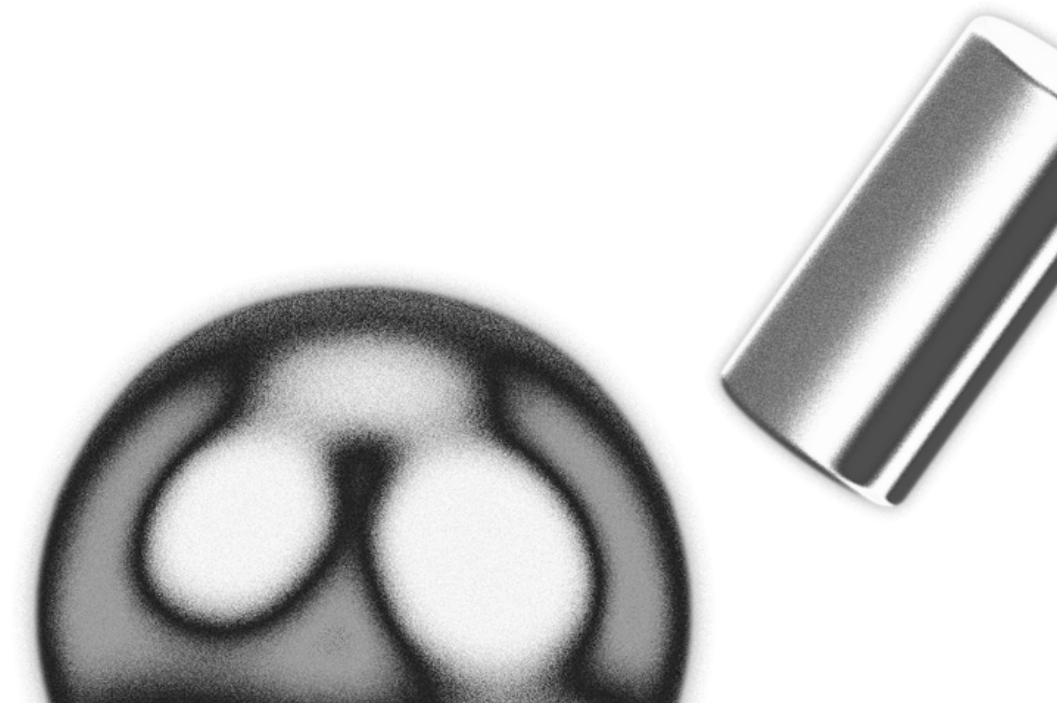


Adding prefix to text for MLM training

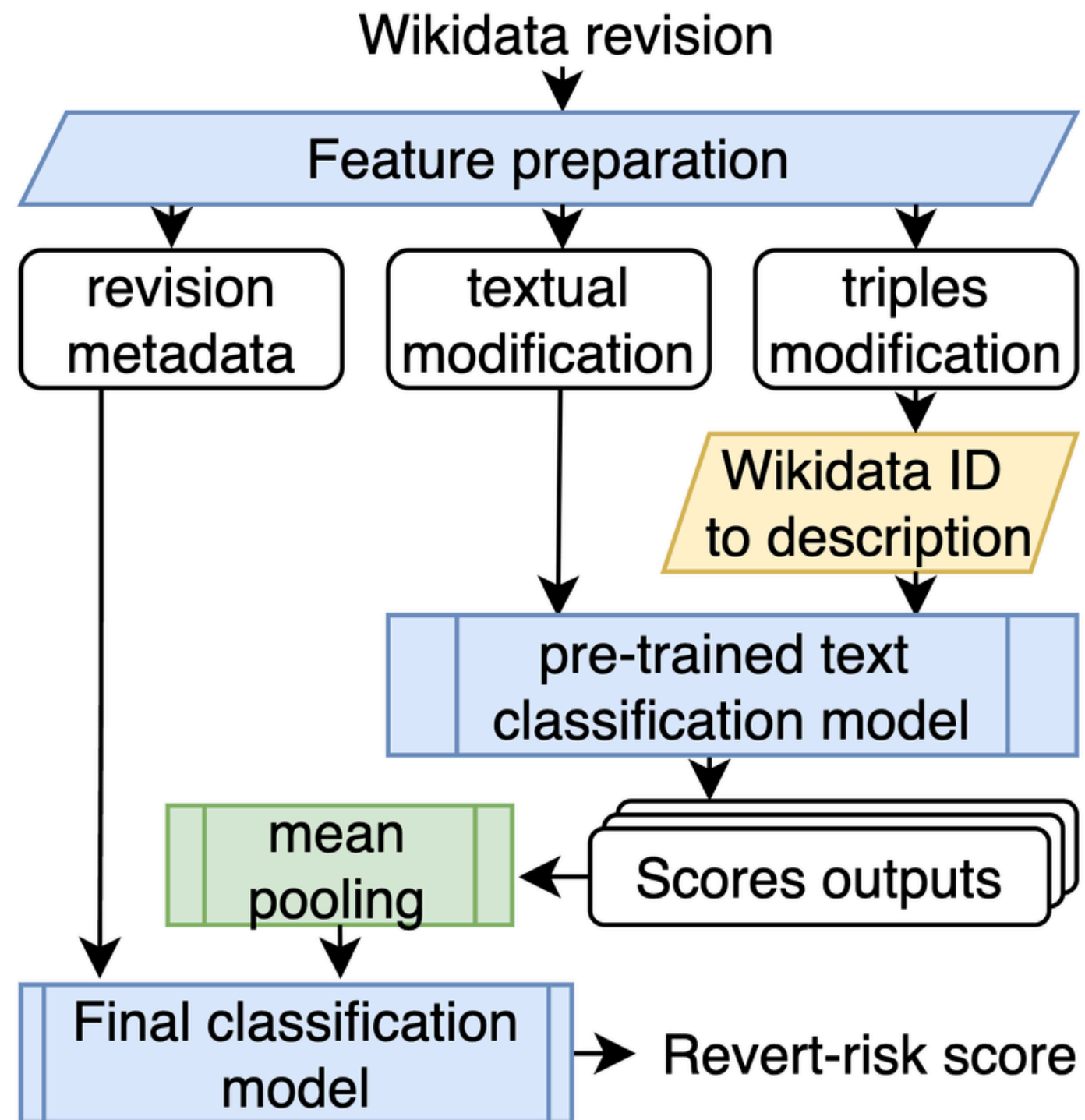
Add: <text>
Remove: <text>
Change: <old text> <SEP> <new text>

Main characteristics of collected data:

- Human-created Wikidata revisions
- Dated from 01-01-2021 to 01-01-2024
- All reverted revisions + random sample of five times as many unreverted revisions.
- Deepdiff parsed differences of content in the form of inserts, removes, and changes
- Graph changes are mapped in the text
- Adding prefix of change type to tune the single MLM for all changes types.



System design



- **Revision metadata:**
 - Passing features directly to the final classifier.
- **Graph-based changes:**
 - Mapping Wikidata IDs to their corresponding English labels.
 - Continue the same procedure as with the text text-based changes.
- **Text-based changes:**
 - Add a prefix corresponding to the type of edit (add, remove, change).
 - Get the output from the pretrained MLM with a classification head.
 - Apply mean and max pooling to the list of scores of each signal to extract the final unified feature set.
- **Final Classification Model**
 - Pass the features to the final classifier.

Metrics

Table 1: System performance on holdout testing set.

Model	AUC	Pr@R0.99	Pr@R0.9	Pr@R0.5
Rule-based	0.629	0.173	0.173	0.173
ORES	0.687	0.175	0.198	0.328
Graph2Text	0.846	0.209	0.304	0.572

Table 2: Fairness metrics evaluation.

Model	DIR	AUC diff
Rule-based	-	-
ORES	5.71	-0.092
Graph2Text	2.32	-0.041

Disparate Impact Ratio (DIR)

$$DIR = \frac{Pr(\hat{Y}=1|D=unprivileged)}{Pr(\hat{Y}=1|D=privileged)}$$

Pr - probability

$\hat{Y}$ - predicted value,

D - a group of users (anon. or registered)

AUC diff - difference between AUC scores of an unprivileged group (anon. users) and privileged (registered users)

Future work

- **Improve parsing of changes to Wikidata**
 - e.g., include qualifier changes to parsing
- **Enhance the process of converting graphs to text**
 - e.g., experiment with alternatives to English label mapping
- **Use advanced data filtering techniques**
 - e.g., address anomalies like self-revert or edit-wars
- **Experiment with alternative modeling techniques**



Thank you!

 mykola.trokhymovych@upf.edu

 diego@wikimedia.org